

# Zach Zukowski

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## STABLECOIN OPERATIONAL RISK · SCENARIO MODELING · TOKEN-ECONOMIC MODELING

- 74-vector risk taxonomy · 64 testable controls · 624-scenario simulation battery · Thorchain \$200M insolvency modeled 13 months ahead
  - 8 SSRN papers · 14 federal comment letters
- 16 USPTO patent applications across 8 families (Canonical-State Architecture for multi-agent AI research)

## SUMMARY

Operational and token-economic risk modeler who turns scenario simulation and coalition analysis into pre-failure risk calls and no-exposure positions, from protocol insolvency to governance capture. Founder of Tokenization Systems; first hire at Borderless Capital, an RIA.

Focus: operational risk, token-economic modeling, scenario modeling, stablecoin and tokenization risk · Monte Carlo, difference-in-differences, PID controller design, supply-equilibrium and coalition-power analysis · OCC, FDIC, FinCEN, Treasury GENIUS, MiCA.

## EXPERIENCE

### Founder · Tokenization Systems

*Feb 2026 to Present*

- Authored an eight-paper SSRN-live research program covering the [Control Layer Intensity Index](#) (CLII) for gateway concentration, the [Minimum Viable Equivalence Pack](#) (MVEP) for institutional tokenization diligence, programmable monetary-policy design ([Adaptive Tokenomics](#)), [governance concentration](#), and [operational risk in token economies](#). Three additional papers in development (B5 Bankruptcy-Priority Misaligned Incentives, A7 Control Layer Theory, SVB Reserve Concentration); methodology paper in preparation.
- Contributed fourteen comment letters across a federal regulatory program (OCC bank stablecoin engagement [RIN 1557-AF41](#) plus [supplemental control-layer extension](#); Treasury GENIUS Act state similarity; FDIC payment-stablecoin reserve standards [RIN 3064-AG19](#) and stablecoin-subsidiary issuance approvals [RIN 3064-AG20](#); FinCEN AML/CFT [FINCEN-2026-0034](#); FinCEN-OFAC PPSI sanctions [FINCEN-2026-0100](#); joint banking AML/CFT [91 FR 18304](#)); [Routing the Dollar](#) prepared for the Federal Reserve Fifth Conference on the Dollar's International Roles (Jun 2026); [Governance Concentration](#) accepted for publication at Frontiers in Blockchain.
- Active dialogues across U.S. national-security research, major securities exchanges, central securities depositories, global payments messaging infrastructure, tier-1 bank tokenization platforms, and SEC-registered transfer agents on stablecoin, deposit-token, tokenized-deposit, tokenized money-market-fund, tokenized-commodity, and tokenized-equity deployment.

### Senior Investment Analyst, Digital Assets · Borderless Capital (RIA; first hire)

*Dec 2019 to Feb 2026*

#### Risk Detection & Loss Prevention

- Predicted Thorchain's **\$200M insolvency 13 months in advance**, leading to a no-exposure posture before the predicted failure mode materialized.
- Prevented hostile governance capture on a top-25 blockchain via voting-path simulation under coalition-attack scenarios; surfaced the vulnerability and mitigation route before exploitation.
- Identified ~15% misreporting of circulating supply in a top-25 Layer 1 protocol by independently modeling its tokenomics (emission schedule, vesting, burn dynamics); the discrepancy had propagated to CoinMarketCap and CoinGecko, and was corrected after partnering with the protocol's foundation engineers.

#### Controls & Risk Frameworks

- Built operational-risk taxonomies for tokenized systems: 74 risk vectors mapped to 64 testable controls and 31 program objectives across a 624-scenario simulation battery (published as [Operational Risk in Token Economies](#)); Lido V3 staking-vault production case study sets operator collateral by risk-score alignment.
- Built apples-to-apples multiple on NAV (mNAV) methodology for Digital Asset Treasury Stocks (DATs) and shipped a daily mNAV dashboard sourcing SEC filings and press releases; standardized treatment of non-crypto assets in the numerator across issuers (issuer-reported mNAV figures use inconsistent conventions that undermine cross-DAT comparison).

#### Diligence & Advisory

- Translated regulatory change (GENIUS, CLARITY, Singapore MAS, MiCA, state money-transmitter and trust-charter regimes) into product, reserve, routing, and compliance decisions across U.S., Singapore, Cayman / BVI / Bermuda, and EU; coordinated cross-functional partner work spanning Product / Engineering, Risk (operational-risk taxonomy), and Legal.
- Mapped jurisdictional tax / regulatory advantages of DAT structures over direct crypto holding: Japan's Metaplanet DAT cuts crypto capital-gains tax from up to 55% to 20% (equity rate, further reducible via Japan's IRA structure); Hong Kong's Southbound Connect could give mainland Chinese capital regulated access to crypto banned domestically.
- Advised 100 portfolio companies 1-on-1 (50+ multi-session) on token mechanism design (stablecoin reserves, governance-token allocation and incentive design, yield wrappers, burn/emission dynamics); standards defined in [Introduction to Tokenomics](#).
- Built quantitative token-economic models forecasting emissions, fee capture, subsidy-to-revenue trajectories, and governance concentration under wide scenario ranges; supply-equilibrium and slashing dynamics supporting Layer 1 mechanism-design advisory.

## SELECTED RESEARCH

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- [Operational Risk in Token Economies](#). Operational-risk taxonomy (74 risk vectors, 64 testable controls, 31 program objectives) mapped to a 624-scenario battery; Lido V3 staking-vault case study.
- [Adaptive Tokenomics](#). Industrial control engineering (PID, Proportional-Integral-Derivative) applied to token-supply emission as programmable monetary-policy mechanism; 806-run agent-based simulation battery; PID emission control cuts worst-case node-count variance from CV (coefficient of variation) 0.544 (static emission) to 0.087 (PID), functioning as downside insurance under competitor-shock attack.
- [Routing the Dollar](#). Prepared for Federal Reserve / FRBNY Fifth Conference on the Dollar's International Roles (Jun 2026). Gateway concentration; CLII across 19 gateways with Tier 1 a 56% / 44% Coinbase-Circle duopoly; on Ethereum, transfer volume doubled while unique counterparties fell 25% and cross-gateway connections dropped from 5.8% to 2.6%.
- [Who Controls Agentic Payments?](#) First control-layer measurement of agentic payments (x402). May 2026 facilitator concentration: Base HHI 6,009 by count and 4,040 by dollar volume; Solana 4,203 and 4,915, all above the 2,500 highly-concentrated threshold at under 14 months of market age.
- [FSB AI Sound Practices response](#). Filed 2026-07-01; the program's first international standard-setter filing. Three practice additions (composition-chain documentation, execution-authority evidence, dependency-concentration monitoring) plus the invited nonbank case study on settlement facilitators.
- [Minimum Viable Equivalence Pack](#). Ten-category institutional diligence standard; finds bank-chartered stablecoin issuers face ~22x the regulatory capital requirements of money-transmitter-chartered issuers at \$40B scale; production-validated by Circle's BUIDL March 27, 2025 incident (23 hours of limited availability). Under review at the Journal of Financial Regulation (editor-invited).
- [Three Tokenization Strategies](#). Each of three strategies satisfies different MVEP categories with measurably different strength (Regulated Collateral Circuit, Multi-Issuer Yield Stack, Tokenized Equity Rails); Securitize as cross-strategy concentration risk.
- [Seven Dollars](#). Cross-product framework for major dollar-denominated digital products; CLII / MVEP decoupling; stress propagation through dependency chains (SVB, Circle pool, Paxos).
- [Dollar v3](#). Four digital-dollar objects (payment stablecoins, tokenized deposits, deposit tokens, yield wrappers); typology for inheritance of control environments, backstop expectations, and failure modes; load-bearing for institutional scoping of product shape and counterparty topology.
- [Auditing Governance Concentration Beyond Token Allocation](#). Accepted for publication at Frontiers in Blockchain. Live-governance audit of 52 token protocols: launch allocation does not predict governance concentration; DePIN governance is more concentrated than DeFi after correction (Cohen's  $d = 0.65$ , a directionally robust medium effect).
- [Who Burns the Tokens?](#) Burn-mechanism design (carrier contracts vs. subscriptions) drives demand concentration: Helium's carrier-contract model concentrates ~90% of token burns in 5 purchasers (HHI 0.22) despite ~1M hotspots; coalition-power analysis extends standard concentration measures for single-address risk.

## TECHNICAL

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**Methods:** policy-shock impact estimation (difference-in-differences), Monte Carlo simulation, PID controller design (industrial control engineering), coalition-power analysis, supply equilibrium modeling, build-versus-partner framework design

**Data:** Dune Analytics, Nansen Pro, Artemis, DefiLlama, CoinGecko, Helius (Solana), Etherscan, Blockscout (EVM), Spacescope, Token Terminal, Tally (DAO governance), rated.network, Beaconcha.in (validator data), FRED (macro); automated SEC filing and press release monitoring

**Stack:** Python (pandas, statsmodels, scipy, matplotlib), R (fixest, did, synthdid), SQL (DuneSQL, Trino), JavaScript / TypeScript; 22 MCP and API connections wrapping the data sources above plus GitHub; Claude.ai + Claude Code + Cursor with 36-skill library and 17-file living-knowledge architecture; n8n, Apify, Git / GitHub

## RECOGNITION

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- Uniswap Foundation governance committee (2025)
- ETHDenver pitch judge (2025)
- [Introduction to Tokenomics](#) article publicly endorsed by Binance founder CZ on day of publication

## FRAMEWORKS AUTHORED

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- Control Layer Intensity Index ([CLII](#)), Minimum Viable Equivalence Pack ([MVEP](#)), MVEP-gated Conditional Pass-Through ([MGCPT](#); [AG19 Rec 7](#)), Three-Category Reserve Disclosure ([3CRD](#)).

## EDUCATION

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**High Point University** | B.S.B.A., Business Administration (Magna Cum Laude) · Investment Club President  
· Lingnan University, Hong Kong (exchange)

2018